01 / COMP5331 / RESEARCH PROJECT

Beyond Random Anomalies

Graph anomaly detection under ranked, approximately matched injections

Completed experiments on MovieLens 100K and UCI Messages
September 2026 | Reproducible experimental report

Research question

Do simple graph features remain sufficient when synthetic anomalous links are made closer to normal interactions in endpoint activity, local structure and temporal context? This study treats the injection procedure as an experimental variable and preserves a separate, faithful reproduction of the released TGF baseline [1, 2].

Dataset	G0 / RF TGF	G3 / RF TGF	G3 / RF extended	G3 - G0 / TGF
MovieLens 100K	0.830	0.372	0.691	-0.458
UCI Messages	0.599	0.357	0.537	-0.242

Table 1. Mean test average precision over five paired injection/model seeds. G0 is a random legal non-edge; G3 applies endpoint, local and temporal candidate rankings. AP is a ranking metric, not classification accuracy.

The injection procedure changes measured difficulty

RF with TGF loses 0.458 AP on MovieLens and 0.242 on UCI from G0 to G3. Additional local and temporal features recover part of the G3 performance. The generators nevertheless retain substantial distribution gaps, and UCI contains an unmatched repeated-pair shortcut. These results support sensitivity to benchmark construction, rather than a claim of realistic fraud detection.

Coverage: 840 main evaluations, 200 sensitivity evaluations and 40 prior-pair feature ablations; plus 20 official-protocol/timing fits. All numerical tables and figures in this document are loaded from saved outputs.

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Data and temporal evaluation

Property	MovieLens 100K	UCI Messages
Observed events	100,000	59,795
Nodes	1,682 movies + 943 users	1,899 users
Unique observed pairs	100,000	13,838
Graph type	Bipartite, movie then user	Undirected, repeated events
Distinct timestamps	49,282	58,911

The complete cleaned examples are pinned to the TGF repository commit. UCI raw input contains 59,835 messages; official cleaning removes duplicated timestamp/pair events. Ratings and message direction are not predictor inputs. Source URLs and SHA-256 values are retained in data/manifest.json [2-4].

Dataset	Train normals	Validation normals	Test normals	Test anomalies / mixed rows
MovieLens 100K	64335	15778	18886	994 / 19880
UCI Messages	53839	3169	1787	94 / 1881

Table 2. Actual main-condition counts after warm-up. Split boundaries are 60% and 80% of elapsed observation time, not event-count quantiles. The late UCI period is sparse, making its test sample much smaller.

Score first, then update the complete timestamp batch

All features for time t are computed from events with timestamps strictly less than t. Normal and synthetic events at t are then inserted into a shared mixed history. H=1000 retains the last 1,000 events; recent activity uses an independent 86,400-second window. Warm-up includes the entire timestamp containing observed event 1,000. Earlier unlabeled test events can update later test history, as in a stream.

The main generator requests round(N_normal x 0.05 / 0.95) anomalies separately within each split. Seeds 11, 23, 37, 51 and 71 share reference timestamp schedules across G0-G3. Main pair exclusion uses the full observed pair set, an explicit offline construction oracle. The feature history itself remains causal.

Model selection uses validation AP only. Selected models are refit on train plus validation before final testing. The source-generator model is fixed when applied to other target generators; target data never selects its configuration.

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Generators, features and classifiers

Condition	Candidate selection rule
G0 / random	Uniformly sample one legal, previously unobserved pair.
G1 / endpoint ranking	Draw 128 legal pairs; retain the 16 nearest reference endpoint log-degree/strength profiles; choose randomly.
G2 / local ranking	From the endpoint 16, retain 4 nearest local profiles; choose randomly.
G3 / temporal ranking	From the local 4, choose the nearest recent-activity/recency profile.

Distances are fixed RMS distances after prescribed transformations. Local and temporal ties use seeded randomness. These are finite-pool nearest-rank controls (128 -> 16 -> 4), not exact matching or hard tolerance guarantees. The chosen streams subsequently have different mixed histories; population-level closeness need not improve monotonically.

Feature group	Definition and scope
TGF (44 / 33)	Active graph summaries; endpoint degrees and strengths; pair weights; equal-value and strictly-greater-value counts. Bipartite / unipartite counts shown.
Local (2 / 3)	MovieLens: simple length-three path count and degree-normalized path count. UCI: common neighbors, Jaccard, Adamic-Adar.
Temporal (7)	Endpoint counts in the prior day, capped log recencies, recent-active flags, and lifetime pair_seen_before.
Extended (53 / 43)	TGF + local + temporal. Ordinary common neighbors are not used across MovieLens partitions because they are identically zero.

Random Forest

100 trees, balanced_subsample class weights, validation choice of min_samples_leaf in {1, 5}, other recorded scikit-learn defaults. Logistic regression uses nonnegative log1p features, a training-fitted StandardScaler, balanced class weights, liblinear and validation choice of C in {0.1, 1, 10}.

The main factorial fits 160 final models (2 datasets x 4 generators x 2 feature sets x 2 classifiers x 5 seeds), each tested on four generators. Five RF feature-group ablations add 200 models. Mean +/- sample SD measures conditional seed variation, not population uncertainty across datasets or independent time folds.

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Faithful official reproduction

Dataset	Feature timing	ROC-AUC	Average precision
MovieLens 100K	Official after arrival	0.9881 ± 0.0004	0.8127 ± 0.0038
MovieLens 100K	Strict prior timestamp	0.9839 ± 0.0007	0.7474 ± 0.0050
UCI Messages	Official after arrival	0.9639 ± 0.0009	0.4894 ± 0.0100
UCI Messages	Strict prior timestamp	0.9628 ± 0.0008	0.4915 ± 0.0034

Table 3. Five RF seeds (42, 0, 1, 2, 3), with a fixed released injection and fixed training split. These SDs describe RF randomness only. The original notebook uses a chronological 70/30 row split, balanced normal downsampling and 100-tree RF; it does not randomly split train and test.

Exact released-feature agreement

The original generator was run without source edits. All 105,000 MovieLens rows and 62,784 UCI rows match the published feature cache after excluding the runtime cost field. Original learning cells execute unchanged; seed-42 ROC-AUC reproduces the notebook values 0.9880 and 0.9635. The first 1,000 featureless rows are removed, leaving 44 and 33 predictors.

Raw preprocessing and injection also executed

All original raw-data preprocessing notebook cells were executed in isolated output directories. Both cleaned files are byte-identical to the committed versions. Original shell injection and checking passed for 100,000 + 5,000 MovieLens events and 59,795 + 2,989 UCI events. Git-blob LF files and a Windows stdout newline adapter preserve the original algorithm. The unseeded fresh injection is retained but does not replace the fixed stream used for faithful metric comparison.

Observation-time sensitivity

Official features include the current event and earlier rows at the same timestamp. The paired alternative freezes each timestamp group and keeps the same event stream, row split, training indices and RF settings. MovieLens AP falls by 0.0653; UCI changes by +0.0021. This contrast changes both current-event and tied-event visibility. It is not evidence of label leakage, nor does it explain all random-injection discrimination.

Official 5% means anomalies / normals, giving about 4.762% mixed prevalence. Its injection excludes duplicate triples, whereas the main causal benchmark excludes globally observed pairs and uses 5% mixed prevalence. Therefore their scores are separate protocol results, not a direct method ranking.

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Main result: injection changes difficulty

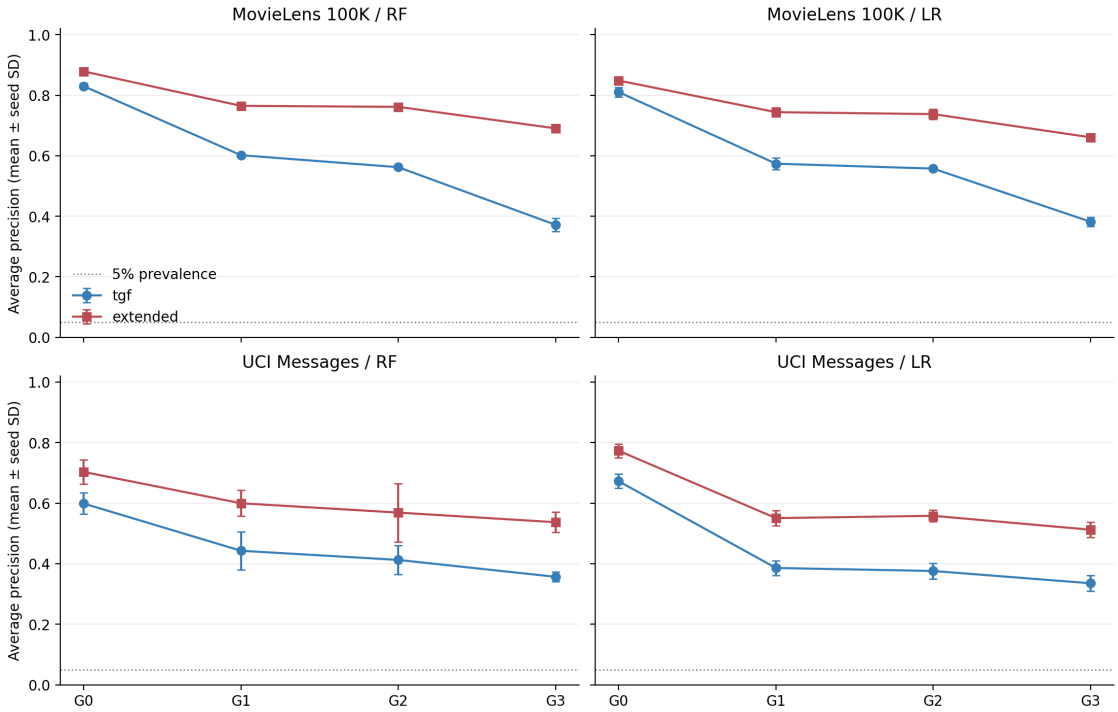


Figure 1. Diagonal tests train and test on the same generator. Error bars are sample SD over five seeds. The dotted 5% line marks main-condition prevalence, the random-ranking reference.

Dataset	Gen.	RF / TGF	RF / extended	LR / TGF	LR / extended
MovieLens 100K	G0	0.830 ± 0.009	0.879 ± 0.007	0.810 ± 0.016	0.848 ± 0.006
	G1	0.602 ± 0.007	0.765 ± 0.011	0.574 ± 0.019	0.744 ± 0.015
	G2	0.562 ± 0.005	0.762 ± 0.009	0.558 ± 0.006	0.738 ± 0.017
	G3	0.372 ± 0.022	0.691 ± 0.014	0.382 ± 0.015	0.661 ± 0.010
UCI Messages	G0	0.599 ± 0.035	0.703 ± 0.040	0.672 ± 0.023	0.773 ± 0.023
	G1	0.443 ± 0.062	0.600 ± 0.043	0.386 ± 0.024	0.551 ± 0.025
	G2	0.413 ± 0.047	0.569 ± 0.096	0.376 ± 0.026	0.558 ± 0.020
	G3	0.357 ± 0.016	0.537 ± 0.034	0.335 ± 0.026	0.512 ± 0.025

Table 4. Average precision, mean +/- sample SD. AP is computed by average_precision_score; trapezoidal PR-AUC is saved separately. Different classifiers and feature groups need not follow identical trends. The full paired per-seed contrasts are supplied with the report tables.

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Cross-generator transfer: RF

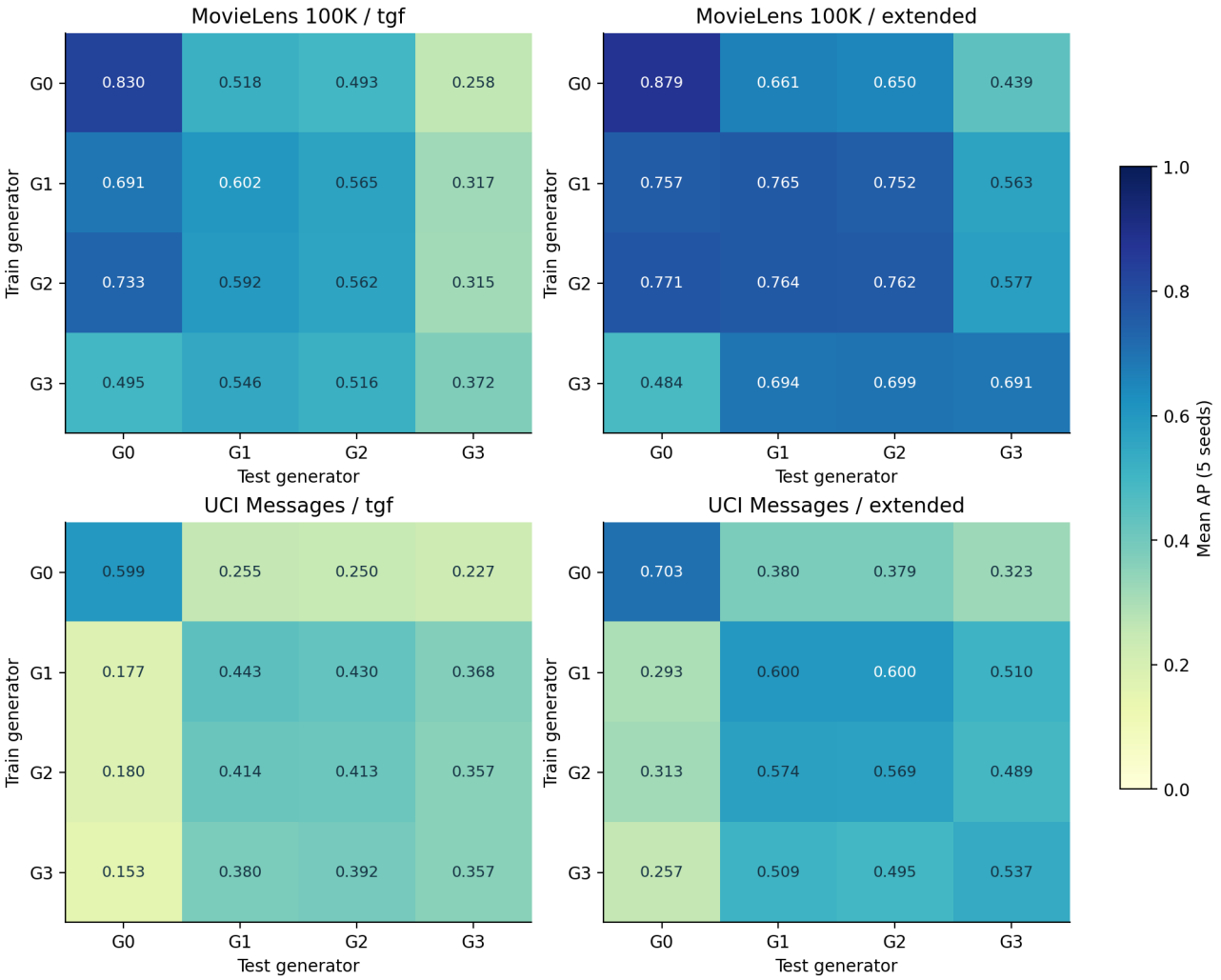


Figure 2. Mean AP over five seeds. Rows identify the training generator; columns identify the test generator. Each row applies one fixed fitted model per seed, so off-diagonal differences are evaluation-distribution changes, not target-specific retraining.

Dataset	Features	G0 -> G0	G0 -> G3	G3 -> G3
MovieLens 100K	tgf	0.830	0.258	0.372
MovieLens 100K	extended	0.879	0.439	0.691
UCI Messages	tgf	0.599	0.227	0.357
UCI Messages	extended	0.703	0.323	0.537

Table 5. Reading one controlled transfer contrast. G0 -> G3 tests a random-injection-trained model on G3; G3 -> G3 also adapts training to G3. The heatmaps make possible asymmetry visible. Numeric annotations keep the matrices interpretable in grayscale.

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Cross-generator transfer: LR

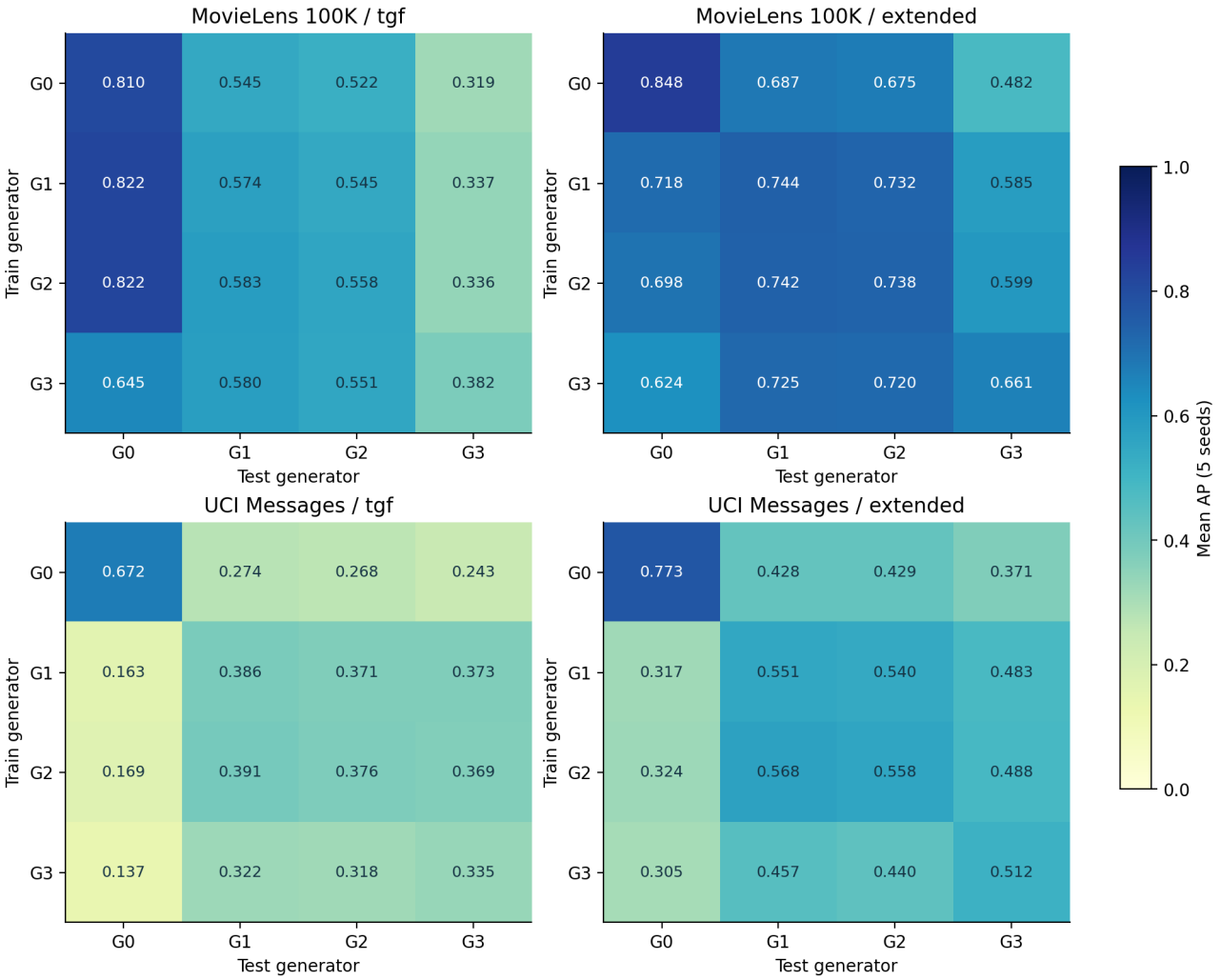


Figure 3. Mean AP over five seeds. Rows identify the training generator; columns identify the test generator. Each row applies one fixed fitted model per seed, so off-diagonal differences are evaluation-distribution changes, not target-specific retraining.

Dataset	Features	G0 -> G0	G0 -> G3	G3 -> G3
MovieLens 100K	tgf	0.810	0.319	0.382
MovieLens 100K	extended	0.848	0.482	0.661
UCI Messages	tgf	0.672	0.243	0.335
UCI Messages	extended	0.773	0.371	0.512

Table 6. Reading one controlled transfer contrast. G0 -> G3 tests a random-injection-trained model on G3; G3 -> G3 also adapts training to G3. The heatmaps make possible asymmetry visible. Numeric annotations keep the matrices interpretable in grayscale.

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How closely were anomalies matched?

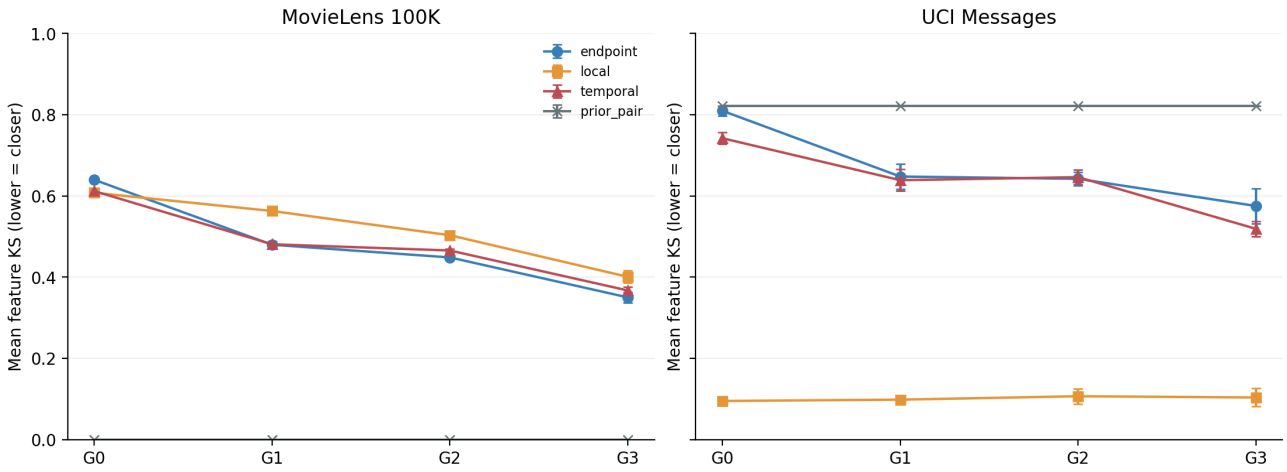


Figure 4. Mean two-sample KS statistic within feature groups, then mean +/- SD across seeds. Lower values indicate smaller marginal normal/anomaly gaps; the prior-pair indicator is shown separately.

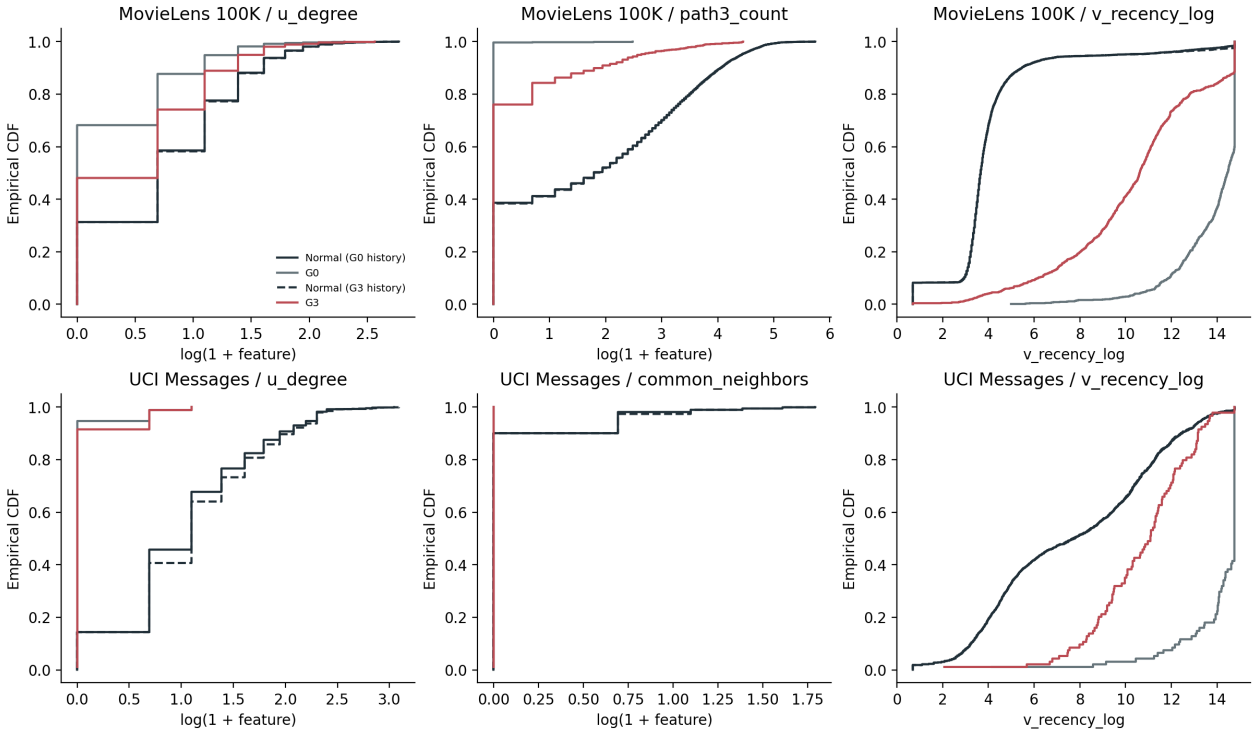


Figure 5. Test empirical CDFs for one fixed seed (11), comparing G0 and G3. Degree and local-count axes use log(1+x); recency is already transformed. Solid/dashed normal curves reflect different mixed histories. CDFs are illustrative, not additional independent replicates.

Matching is incomplete: the fraction of G3 injections whose endpoint RMS log-distance exceeds log(2) is MovieLens 100K 76.3%, UCI Messages 69.1%. This threshold is a diagnostic, not an enforced caliper. Substantial degree/activity separation remains; UCI local marginal gaps need not shrink from G0 to G3. Avoid describing these conditions as perfectly structure-preserving.

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Which features carry the signal?

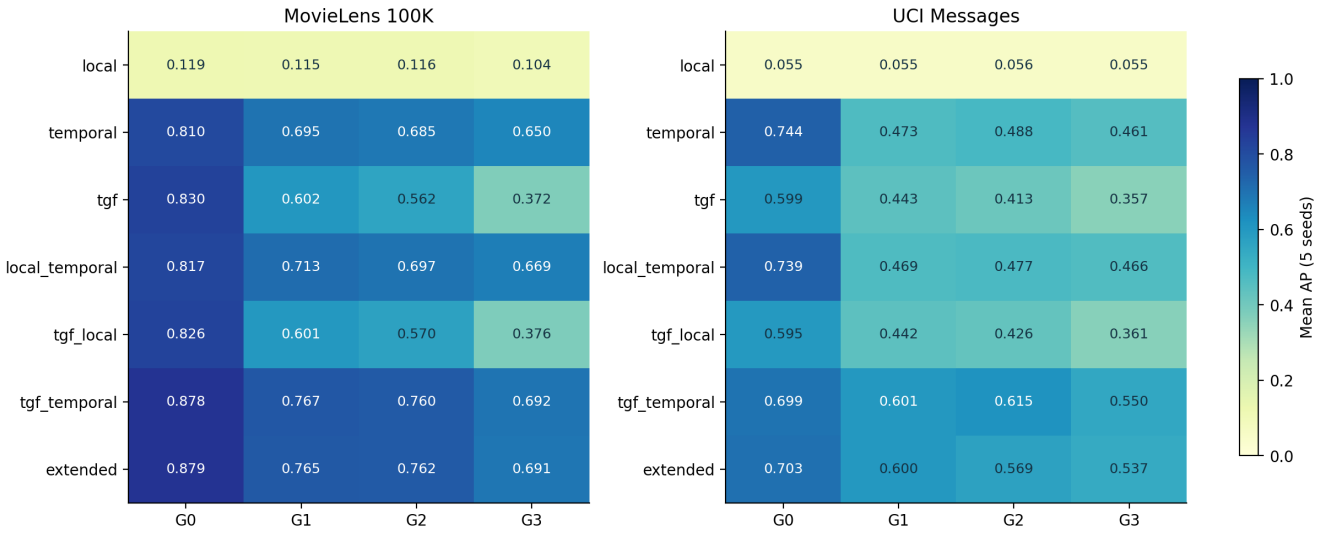


Figure 6. RF feature-group ablations, five-seed mean AP. Local-only, temporal-only and local+temporal test feature sufficiency; TGF+local and TGF+temporal isolate additions to the baseline.

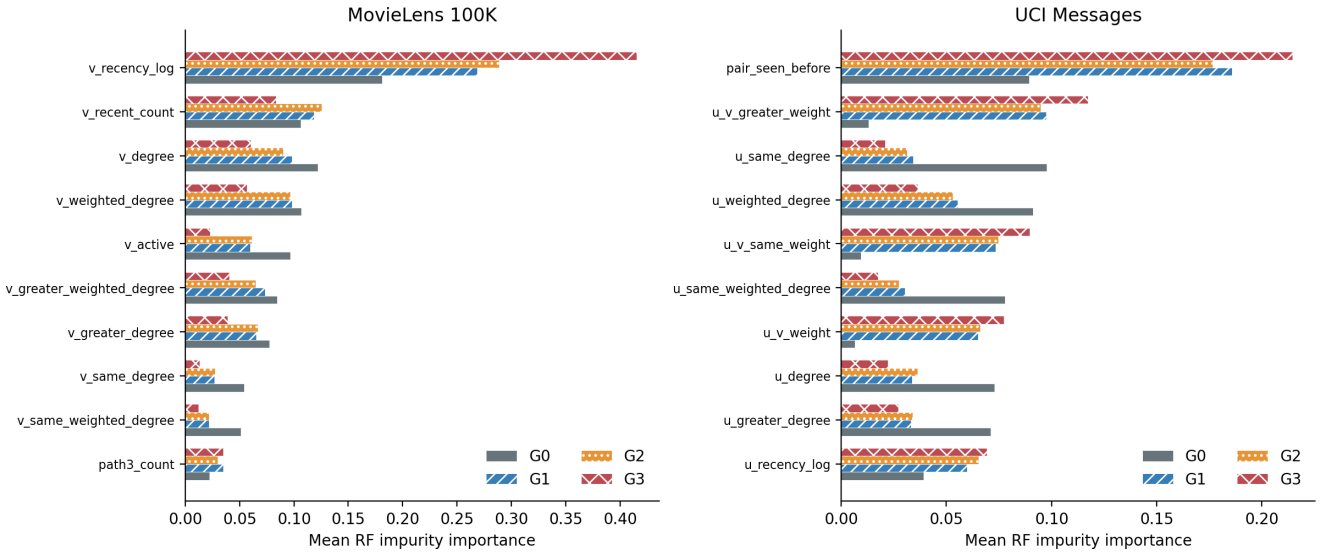


Figure 7. Mean RF impurity importance for the extended feature set. The displayed features are the largest across the generators within each dataset. Importances are descriptive and can be biased by correlated or high-cardinality predictors.

On G3, adding all local and temporal features changes RF AP by MovieLens 100K +0.319, UCI Messages +0.180. This gain is a feature-set comparison under the same seed and generator. UCI interpretation requires the prior-pair controls on page 11 because global non-edge injection makes all synthetic pairs novel.

The G3 group ablations locate most recovery in the temporal group (MovieLens 100K: TGF+temporal 0.692, TGF+local 0.376; UCI Messages: TGF+temporal 0.550, TGF+local 0.361). Local-only performance is much lower. Adding all features need not outperform TGF+temporal.

Neither feature importance nor group ablation is a causal attribution. Correlated endpoint, pair-weight and recency statistics can substitute for one another. The benchmark supports an empirical comparison of available signals, not identification of a unique mechanism.

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Sensitivity to prevalence and history

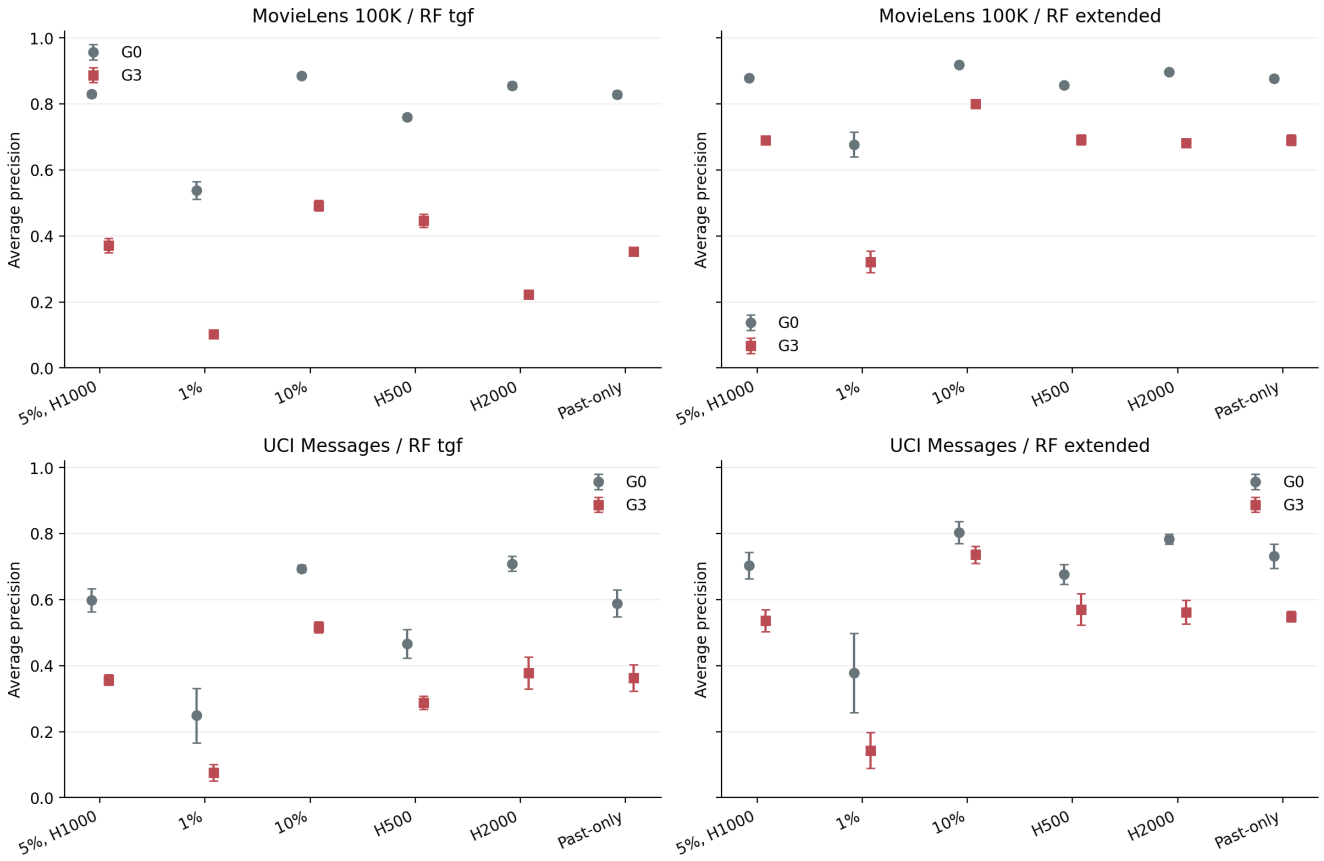


Figure 8. Each condition changes one main-protocol parameter: mixed anomaly fraction 1% or 10%, H=500 or 2,000, or past-only pair exclusion. RF, G0/G3 and both feature sets are repeated over all five seeds on both datasets.

Dataset	Features	Main G3	H500 G3	H2000 G3	Past-only G3
MovieLens 100K	tgf	0.372	0.446	0.223	0.353
MovieLens 100K	extended	0.691	0.691	0.681	0.691
UCI Messages	tgf	0.357	0.288	0.378	0.363
UCI Messages	extended	0.537	0.570	0.562	0.549

Table 7. Selected G3 mean AP sensitivity values. Past-only exclusion removes the full-future-pair absence constraint but still excludes current-timestamp observed duplicates. The full sensitivity CSV also records realized prevalence and ROC-AUC.

AP depends on prevalence. Directly comparing AP at 1%, 5% and 10% mixes prevalence with detector ranking changes; those values are not a pure difficulty scale. History-size and absence sensitivities use the main prevalence. Fixed elapsed-time splits remain unchanged, so UCI late-period sparsity persists.

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Residual novelty shortcut and controls

In the UCI test set, 82.09% of normal events have a pair previously observed in lifetime mixed history, while globally absent injected pairs are all novel. G1-G3 do not match this property. MovieLens has no corresponding repeated-normal-pair signal. This constraint can make temporal features useful without showing that recency itself recovers performance.

Dataset	Gen.	Full AP	Novel-normal AP	Filtered prevalence	Filtered n
MovieLens 100K	G0	0.879	0.879	5.0%	19880
MovieLens 100K	G3	0.691	0.691	5.0%	19880
UCI Messages	G0	0.703	0.707	22.7%	414
UCI Messages	G3	0.537	0.537	22.7%	414

Table 8. Existing RF extended predictions are reevaluated after removing repeated normal pairs. Models are not retrained. The filtered class prior changes, so a higher or lower filtered AP cannot be attributed only to difficulty. Counts and prevalence are reported alongside the metric.

Dataset	Gen.	Extended AP	Without prior-pair AP	Paired change
MovieLens 100K	G0	0.879 ± 0.007	0.877 ± 0.007	-0.001
	G1	0.765 ± 0.011	0.767 ± 0.012	+0.002
	G2	0.762 ± 0.009	0.765 ± 0.008	+0.003
	G3	0.691 ± 0.014	0.697 ± 0.016	+0.006
UCI Messages	G0	0.703 ± 0.040	0.627 ± 0.042	-0.076
	G1	0.600 ± 0.043	0.515 ± 0.056	-0.084
	G2	0.569 ± 0.096	0.477 ± 0.084	-0.092
	G3	0.537 ± 0.034	0.459 ± 0.046	-0.078

Table 9. Forty additional fits remove only pair_seen_before. Each reuses the corresponding extended model's selected min_samples_leaf, refits on train+validation, and applies the same test split. This is a controlled feature deletion, not independently retuned feature selection. TGF still contains recent pair weight, so other novelty proxies remain.

For UCI G3, removing the explicit prior-pair bit changes AP by -0.078, confirming that this feature materially contributes under global non-edge injection. MovieLens's bit is constant zero; its small score changes reflect RF refitting/feature-sampling effects rather than removal of a novelty signal.

Within the UCI G3 novel-normal subset, full and no-pair AP are 0.537 and 0.541. The small change there, compared with the full-set drop, supports a role in separating repeated normal pairs from novel injected pairs. It does not establish better ranking of new normal interactions.

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Reproducibility and verification

Experiment block	Final fitted models	Test evaluations
Core RF/LR x TGF/extended	160	640
RF feature-group ablations	200	200
Five sensitivity conditions	200	200
Remove pair_seen_before	40	40
Causal-extension total	600	1080
Separate official/timing reproduction	20	20

Hyperparameter candidate fits are additional internal validation fits, not counted as final models. The 140 cached generation conditions comprise 40 main streams and 100 sensitivity streams. Every scored event is retained; these experiments do not downsample normal observations in the causal extension.

Verification performed

Independent tests compare original TGF formulas, active-population ranks and medians, history eviction, separate recency windows, timestamp-batch invariance, bipartite paths and legality. Generator audits verify requested class counts, unique legal injected pairs, strictly prior maximum history timestamps and reproducible cache fingerprints. Model-selection tests check source-only validation and feature alignment.

The official sensitivity outputs pass independent score-level verification: 20 prediction files, 497,350 scored rows, recomputed AP and ROC-AUC within 1e-12, paired labels and row indices equal. Fresh preprocessing has byte-for-byte source equality and original injection checks pass. Saved failures are excluded until successfully rerun; they are not substituted by fabricated scores.

Metric records also contain ROC-AUC, trapezoidal PR-AUC, Precision@K and Recall@K. K is the true test anomaly count, so precision and recall at K coincide; these are retrospective ranking summaries, not a deployment threshold selected without labels.

Artifact map

Location	Contents
data/manifest.json	Dataset sources, hashes, split boundaries and counts
results/cache/	140 manifests, complete feature arrays and injection audit logs
results/evaluation/	Fitted models, per-event scores, selection metadata and metrics
results/no_pair/	Forty prior-pair deletion fits and predictions
results/official/	Original-source reproduction, cached-feature checks and fresh pipeline
results/report/	Regenerated figures, tables, summaries and per-seed contrasts
docs/experiment_plan.md	Protocol details and deviations required for interpretation

Regenerate tables/figures: python scripts/report_results.py. Build this PDF using the bundled reportlab runtime: python scripts/build_pdf_report.py. Training/generation entry points are scripts/build_experiments.py and scripts/evaluate_experiments.py; saved manifests fix parameters and seeds.

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Conclusions, limits and references

What the experiments establish

The released TGF example results are reproducible. Within a fixed causal experiment, progressively ranked injections can substantially reduce TGF AP, and inexpensive additional features can recover some performance. Cross-generator testing makes transfer sensitivity visible; a detector trained on one injection distribution need not behave equivalently on another.

What they do not establish

Synthetic labels identify generated edges, not confirmed real anomalies. Matching is approximate and finite-pool; G3 still has large endpoint/activity gaps, while UCI local gaps are not uniformly reduced. Global non-edge exclusion uses future observed pairs and leaves repeated-normal-pair signals unmatched. The past-only and prior-pair controls narrow these issues but do not remove every correlated novelty proxy.

There are only two datasets and one elapsed-time partition per dataset. UCI has few late test events; seed SD does not describe temporal-fold or deployment uncertainty. AP across changed priors is not directly comparable. The original-reproduction protocol differs from the causal extension in injection, feature timing, model weighting and splitting. Optional neural baselines, MIDAS, coordinated anomalies and real labeled fraud are outside this completed single-edge study.

References and primary sources

[1] M. Latapy and S. Rajeh. Trivial Graph Features and Classical Learning are Enough to Detect Random Anomalies. IEEE ICDM, 2025, pp. 1330-1339. DOI: 10.1109/ICDM65498.2025.00142. Publisher record.

[2] Authors' TGF repository, pinned commit af9d575e07c7bc3fd0b4a0b8d0249e7cdd2eb88b. Source code and example datasets; project website.

[3] GroupLens Research. MovieLens 100K dataset. Original dataset page. This report uses the pinned TGF/KONECT cleaned copy.

[4] T. Opsahl and P. Panzarasa. Clustering in Weighted Networks. Social Networks 31(2), 2009, pp. 155-163. UCI messages are distributed as opsahl-ucsocial in KONECT and the pinned TGF repository.

[5] Project artifacts: kdd_proposal.pdf (task specification); docs/experiment_plan.md (fixed protocol); docs/official_audit.md, docs/generator_audit.md and docs/features.md (implementation audits). All report numbers derive from saved CSV/JSON/NPZ experiment outputs.

Metric convention: AP denotes non-interpolated average precision. Mean +/- SD uses sample SD over the stated five seeds. All feature importance and distribution diagnostics are descriptive.